THEIA Preliminary User Manual

Team EV · CptS 484 Project THEIA · Version 1.1 · October 17, 2026

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This manual is written to be read aloud by a screen reader. Every step says what you will hear or feel. You never need to look at the screen.

1. About THEIA

THEIA is a smartphone app that helps blind and visually impaired users find their way inside campus buildings. It uses voice commands, spoken directions, and vibrations to guide you to your destination and to help you when you are lost.

THEIA works alongside your cane or guide dog. It does not replace them.

What you need:

- A smartphone with Bluetooth turned on.
- A campus building that THEIA has a map for.
- Earphones, so you can hear instructions clearly in noisy hallways (recommended).

How you control THEIA:

You can control THEIA by voice or by simple touch gestures. Every action works both ways, so if a hallway is too noisy for voice commands, you can use a gesture instead. THEIA answers every action with speech and a vibration.

To start, open THEIA with your phone's voice assistant (for example, "Hey Siri, open THEIA" or "Hey Google, open THEIA"). This works from the lock screen. Building maps are stored on your phone, so navigation works without an internet connection.

2. First-Time Setup (Caretaker)

A caretaker should set up THEIA with the user before the first trip.

1. Open THEIA and go to Screen 5, Settings.
2. Choose caretaker settings and create a PIN.
3. Add emergency contacts, with their names and phone numbers.
4. Choose whether to turn on automatic fall detection.
5. Choose a default navigation mode: Fast, Safe, or Comfortable.
6. Turn on "No stairs" if the user should not be routed over stairs.
7. Calibrate the user's step length: the user walks a short, measured distance while THEIA counts the steps. If you skip this, THEIA uses an average step of 0.7 meters.
8. Optionally, add the user's class schedule and favorite places.

The settings in this section are safety settings. Only the caretaker can change them, and they are protected by the PIN. The user can change comfort settings, such as volume and speech speed, at any time by voice (see Section 7).

3. Basic Controls

- **Start talking:** tap anywhere on the Voice Home screen, then speak.
- **Confirm:** say “Yes” or double-tap. You will feel one short vibration.
- **Hear the last instruction again:** say “Repeat” or double-tap.
- **Find out where you are:** say “Where am I?” or double-tap with two fingers. THEIA tells you the building, floor, nearest room, and which way you are facing.
- **Get help:** say “Help” or hold two fingers on the screen for 2 seconds.
- **Stop something:** say “Cancel.”

4. Going to a Room (Scenario 1: Stevie)

1. Open THEIA. The Voice Home screen asks, “Where would you like to go?”
2. Tap anywhere on the screen and say your destination, for example, “Take me to Spark 233.”
3. THEIA starts from where you are now, using the nearest location beacon. To start from a different place, say “Start from” and the place name.
4. THEIA repeats your destination and asks, “Spark 233. Is that right?”
5. Say “Yes” or double-tap to confirm. You will feel one short vibration.
6. If there is more than one route, THEIA reads up to three options. Say the number of the one you want. If you don’t answer within 10 seconds, THEIA picks the best route for your current mode.
7. Follow THEIA’s spoken directions, for example, “Walk about 20 steps, then turn left.”
8. Five steps before each turn, THEIA says, “In 5 steps, turn left.”
9. At the turn, THEIA says, “Stop. Turn left now.” You will feel one short vibration.
10. If there is a known obstacle ahead, THEIA warns you 5 steps before it, for example, “Bench on your right in 5 steps, keep left.”
11. When you arrive, THEIA says, “Spark 233 is on your right.” You will feel one long vibration.

THEIA only warns about obstacles that are recorded in the building map, such as benches, pillars, or construction barriers. Keep using your cane or guide dog for people and other moving obstacles.

5. Changing Floors by Elevator (Scenario 3: Mark)

1. Open THEIA and say, “THEIA, lead me to TFLO 131 via elevator.”
2. THEIA only offers routes that use the elevator. The first route it offers is the shortest accessible route with the fewest turns. Say the number of the route you want.
3. Follow the spoken directions to the elevator.
4. When you reach the elevator, you will feel one short vibration and THEIA says, “Call button on your right. Go to floor 1.”
5. Press the call button and enter the elevator when it is safe.
6. Select floor 1 using the elevator buttons.

7. When you leave the elevator, THEIA checks which floor you are on and says, “You are on floor 1.” It then switches to that floor’s map and continues guiding you. If you are on the wrong floor, THEIA tells you.
8. Follow the remaining directions to your destination.

If you miss an instruction, say “Repeat” or double-tap to hear it again.

6. If You Are Lost or Need Help (Scenario 2: Alfred)

This section explains what THEIA does when your way is blocked, when you leave your route, and when you need help from your caretaker or campus police.

6.1 Your Way Is Blocked

1. If construction or another barrier blocks your path, stop walking.
2. Say “This way is blocked.”
3. THEIA says, “OK. Finding a detour.” A new route starts within about 3 seconds.
4. Follow the new directions.
5. If there is no safe detour, THEIA tells you and offers to open Help. Stay where you are until help arrives.

THEIA also sends a report of the blockage to campus accessibility services so the map can be updated.

6.2 THEIA Says “You May Be Off Route”

1. If you miss the next location checkpoint by more than about 30 seconds, or THEIA detects a checkpoint that is not on your route, you will feel three vibrations and hear, “You may be off route. Say 1 to find a new route, or 2 to go back to Spark 233.”
2. Choose one:
    - Say “1,” or tap the top half of the screen, to get a new route from where you are.
    - Say “2,” or tap the bottom half of the screen, to go back to your last known location.
3. Follow THEIA’s directions.
4. If THEIA cannot find any checkpoint nearby, it tells you your last confirmed location and guides you along the nearest wall until it finds one.
5. If you are still lost after 2 minutes, THEIA asks, “Do you want me to text your caretaker your last location?” Say “Yes” or “No.”

Losing your location does not call anyone by itself. THEIA always asks first.

6.3 Asking for Help

1. Say “Help,” or hold two fingers on the screen for 2 seconds.
2. Screen 4, Help, opens and you feel a strong vibration.
3. THEIA asks, “Do you need help? Text caretaker, call caretaker, call campus police, or cancel?”
4. Choose one:
    - **Text caretaker:** sends your caretaker a text with your building, floor, nearest room, the time, and a map link.
    - **Call caretaker:** calls your caretaker.

- **Call campus police:** calls campus police. THEIA reads your location to you so you can tell them.
- **Cancel:** closes Help and returns to what you were doing.

5. If you don't answer within 10 seconds, THEIA texts your caretaker automatically. Say "Cancel" to stop it.
6. If your caretaker doesn't answer the call, or doesn't reply to the text within 2 minutes, THEIA offers to call campus police.

Example: Alfred is heading to the elevator when construction blocks the hallway. He says "This way is blocked" and gets a detour, but later misses a checkpoint. THEIA offers to reroute him or take him back to Spark 233. When he still can't find his way, he says "Help," and THEIA texts his caretaker, Bella, his exact location so she can find him.

6.4 If You Fall

1. If THEIA detects a possible fall, Screen 4, Help (fall check), opens with a strong continuous vibration.
2. THEIA asks, "Are you OK? I will text Bella in 30 seconds. Say Cancel to stop."
3. If you are OK, say "Cancel" or tap Cancel.
4. If you do not respond within 30 seconds, THEIA texts your caretaker your location and then calls your first emergency contact.
5. If your caretaker does not respond, THEIA offers to call campus police.

Fall detection is a proposed prototype feature. Do not rely on it as a guaranteed emergency service.

7. Settings and Navigation Modes (Scenario 4: Elena)

7.1 Changing Your Settings by Voice

You can change comfort settings at any time, even during a trip. THEIA confirms each change out loud, and the change takes effect right away.

Say	What changes
"THEIA, louder" / "THEIA, quieter"	Voice volume ("Volume 80 percent")
"Speak faster" / "Speak slower"	Speech speed (half to double normal speed)
"Give instructions every 10 seconds"	How often THEIA speaks (every 5 to 30 seconds, or only at turns)
"Vibration off" / "Vibration low" / "Vibration high"	Vibration strength

Example: Elena says, "I'm heading to lunch at the cafeteria," and double-taps to confirm. The hallway is noisy, so she says "THEIA, louder" and "Give instructions every 10 seconds." THEIA confirms each change and speaks more often.

7.2 Navigation Modes

Mode	What it does
Fast	Takes the route with the least estimated walking time, counting extra time for turns and elevator waits.
Safe	Avoids stairs and places with known obstacles, and confirms every turn.

Mode	What it does
Comfortable	Uses fewer turns and routes you have taken before, and avoids areas marked busy or noisy, such as the main lobby. It may take up to 25% longer.

To change mode, say “Fast mode,” “Safe mode,” or “Comfortable mode.” Basic safety rules apply in every mode: THEIA never routes you through closed or inaccessible areas. Your caretaker’s default mode is used until you change it.

7.3 Caretaker Settings

Emergency contacts, fall detection, the default navigation mode, and the “No stairs” preference are caretaker settings. They are protected by the caretaker’s PIN, so they cannot be removed or turned off by accident.

7.4 Class Reminders

If your class schedule is set up, THEIA reminds you about 10 minutes before a class, for example, “Your next class is TFLO 131 at 10:10. Start navigation?” Say “Yes” to start or “No” to skip. THEIA never starts a route without your answer.

8. Quick Reference

8.1 Voice Commands

You say	THEIA does
“Take me to ___”	Starts a route
“Start from ___”	Uses a different starting point
“Yes” / “No”	Confirms or declines
“1”, “2”, “3”	Picks a route or an off-route option
“Where am I?”	Tells you the building, floor, nearest room, and which way you face
“Repeat”	Says the last instruction again
“This way is blocked”	Finds a detour
“Help”	Opens the Help screen
“Cancel”	Stops the current action or countdown
“Louder” / “Quieter”, “Speak faster” / “Speak slower”	Changes voice volume and speech speed
“Give instructions every ___ seconds”	Changes how often THEIA speaks
“Fast mode” / “Safe mode” / “Comfortable mode”	Changes the route style (see Section 7.2)

8.2 Touch Gestures

Gesture	Meaning
Tap anywhere (Voice Home)	Start talking
Double-tap	Yes, or repeat the last instruction
Two-finger double-tap	“Where am I?”
Two-finger hold (2 seconds)	Help
Tap top half / bottom half (Off-route screen)	Option 1 / option 2

8.3 Vibrations

Vibration	Meaning
1 short buzz	Turn now, or your action is confirmed, or you reached the elevator
1 long buzz	You have arrived
3 buzzes	You may be off route
Strong continuous buzz	Help screen or fall check opened

9. Limitations of THEIA

- THEIA works only inside campus buildings that have a THEIA map, and in mapped indoor connections between them.
- THEIA does not guide you to things inside a room, such as a chair or desk.
- THEIA does not detect people or other moving obstacles. Keep using your cane or guide dog.
- THEIA guides you to elevators and stairs, but you operate them yourself.
- Emergency calls and texts need a phone signal.
- Directions are only as accurate as the building map and location beacons.
- THEIA is an assistive navigation app. It does not replace normal mobility and safety precautions.

End of Preliminary User Manual.